

When Machines Speak Like Local Peers: Improving Conversational Experiences with Accent-Adaptive Voice Agents

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Abstract

Public voice agents often underperform for accented English, reducing usability and trust. *LocalMATE* is an end-to-end accent-adaptive airport conversational agent that (i) estimates a user’s accent with calibrated confidence, (ii) retrieves grounded answers, and (iii) returns accent-conditioned speech with a synchronized talking-face video, using a confidence-gated fallback when uncertain. Our accent estimator (WavLM-Base-Plus with layer-weighted statistics pooling) reaches 85% speaker-disjoint accuracy on L2-ARCTIC (Indian/Korean/Spanish) and 92.3% on out-of-domain Korean-accented *Speech Accent Archive*. In a within-participant user study (N=20), correct accent adaptation significantly improved confidence and trust ($p < 0.05$) while reducing user repair attempts. Conversely, incorrect adaptation reduced trust and increased friction. A confidence-gated fallback policy mitigated these effects and restored user confidence.

Index Terms: Multimodal conversational agents, Accent classification, Accent-adaptive speech synthesis, Trust and user perception

1. Introduction

Spoken conversational agents are increasingly deployed in public-facing settings, such as kiosks, help desks, and information robots, where users expect smooth, low-effort interaction. In practice, user speech is acoustically diverse, and those with accented English may face issues if systems are optimized for a narrow range of speaking styles. Prior HCI work shows that multilingual users’ needs are often under-considered in voice-agent interaction and that social context can shape whether users feel comfortable speaking in a particular language or accent to an agent [1, 2]. These concerns are intensified by evidence that automatic speech recognition (ASR) errors are not evenly distributed across demographic speech varieties [3, 4]. Such failures can go beyond usability costs: voice assistant errors may be experienced as microaggressive in cross-race interactions and can negatively affect users’ mood and self-perception [5]. Voice assistant failures also influence long-term dependence, with different failure types degrading user trust in distinct ways [6]. Together, these observations motivate designs that better accommodate accent and linguistic diversity.

Accent and dialect cues influence first impressions and perceived identity, competence, and warmth, and can affect user trust in human-agent interaction [7, 8]. Recent work suggests that limited accent support in synthetic voice services can contribute to users feeling excluded, motivating inclusive accent-adaptive speech technologies [9]. At the same time, in-

creased perceptual realism (e.g., highly human-like voices or realistic audiovisual representation) may unintentionally shift social judgments, requiring controlled perceptual evaluations [10, 11, 12].

Despite this progress, we still lack a systematic understanding of how *accent adaptive response generation* should be deployed in end-to-end conversational systems, especially when responses may be delivered as a realistic talking-face video. In particular, it remains unclear how users perceive and prefer (i) human-recorded speech versus (ii) simulated speech designed to match a user’s accent, and whether these judgments change when the same audio is paired with an audiovisual representation.

In this paper, we study the perceptual impact of accent adaptive responses in a multi-modal FAQ dialogue setting. Initially, we ran a pilot study that isolates presentation effects by comparing four conditions: (1) *Source* human-recorded audio (*S-audio*), (2) *Modified* simulated audio designed to match accent (*M-audio*), (3) talking-face video with *S-audio* (*S-video*), and (4) talking-face video with *M-audio* (*M-video*). Using Likert-scale ratings, we measured perceived naturalness, ease of understanding, accent/tone match, audiovisual synchronization, and preference.

In addition to this pilot, we evaluated an end-to-end accent adaptive conversational system, *LocalMATE*, with a user study designed to answer three research questions:

RQ1: Does correct accent adaptation improve trust and confidence?

RQ2: What is the cost of incorrect accent adaptation?

RQ3: Can confidence-gated fallback – fallback triggered when accent-detection confidence is low – mitigate mismatch harm?

Figure 1 summarizes the proposed workflow. This paper makes three contributions: (i) *LocalMATE*, an end-to-end accent-adaptive conversational agent with confidence-gated fallback and multimodal (talking-face) output; (ii) a robust three-class accent classifier for Indian, Spanish, and Korean-accented English evaluated with speaker-disjoint splits and out-of-domain transfer; and (iii) pilot and main user studies that quantify perceptual preferences and downstream social outcomes (trust, confidence, and interaction friction) for accent-adaptive audiovisual responses.

This work investigates whether accent-adaptive response generation improves user experience in conversational systems and how incorrect accent adaptation affects trust. We further evaluate whether a confidence-gated fallback strategy can mitigate the negative impact of accent misclassification.

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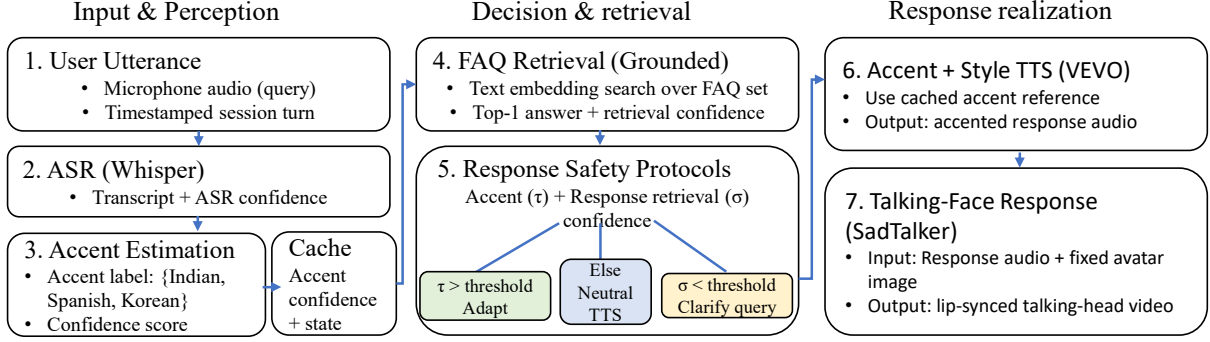


Figure 1: Overall workflow of LocalMATE; our accent-adaptive conversational system. It (i) intakes user speech, (ii) estimates and caches accent with confidence, (iii) transcribes the speech, (iv) retrieves a grounded FAQ response, (v) generates accent conditioned speech with a confidence-gated fallback, and (vi) optionally renders a lip-synchronized talking-face video.

2. Related Work

We review previous work on (1) Accent and social perception in voice interaction, (2) Bias, harms, and repair in voice assistants, and (3) Accent bias in synthetic voice services and multi-modal representation.

2.1. Accent and social perception in voice interaction

Previous research demonstrates that in human-agent interaction, voice design choices heavily influence social judgments and user experience [7]. Furthermore, some studies note that dialect and accent cues can shape perceived competence, warmth, and authenticity, motivating interest in agent speech that is dialect and accent-aligned [8]. Earlier studies on VUI (Voice User Interface) also reported that English regional accents affect user acceptance and preference, highlighting that accent is a prominent factor in voice interfaces [13]. Despite the identified importance of accent aligned responses in a human-centric interaction system, few studies have thoroughly implemented these systems to evaluate their impact on user experience.

2.2. Bias, harms, and repair in voice assistants

ASR systems differ in their demographic performance, where some variants of demographic speech show higher error rates. [3]. These differences can translate into unequal user experiences and affect downstream tasks that need correct speech-to-text translation for aligned responses [14]. Wenzel and Kaufman showed that ASR failures in cross-race voice interactions can be experienced as microaggressive, affecting users’ psychological responses [5]. Another study shows that identical (Automatic Speech Recognition) ASR outputs are perceived differently depending on the speaker’s accent: listeners attribute more blame to non-native speakers and judge them as less credible, even when transcription quality is constant [15]. This highlights that equal model performance does not ensure equal user experience. Voice assistant failures degrade trust and alter future reliance, with mixed-methods work indicating that failure type and context shape trust dynamics [6]. Recent research suggests harm reduction strategies based on communication repair for multicultural voice interactions, highlighting designs that help users recover from recognition errors and mitigate negative downstream impacts [2]. However, comparatively little work has investigated how accent conditioned response strategies such as, producing an incorrect accent versus falling back to a neutral accent under uncertainty, may affect user trust and

confidence in end-to-end spoken systems.

2.3. Accent bias in synthetic voice services and multi-modal representation

Although the quality of synthesis is high, synthetic voice services can neglect certain accents and contribute to digital exclusion [9]. In parallel, perceptual realism and embodiment influence user judgments: prosody and embodiment have been shown to affect perceived naturalness and task outcomes in agent interactions [11], while highly human-like outputs can produce unintended social effects consistent with “unnatural” responses [10]. These findings motivate controlled evaluations of accent-adaptive audiovisual presentations.

3. System

We present *LocalMATE*, an accent-adaptive dialog system for general airport information. Given a user input utterance, the system (i) estimates the user’s English accent using our fine-tuned accent classifier and caches it to maintain a consistent accent state across subsequent conversation, (ii) transcribes the utterance using an ASR model, (iii) retrieves a response from a fixed set of FAQs via embedding-based semantic retrieval, (iv) generates a spoken response conditioned on the user’s accent, and (v) renders a talking-face video synchronized to the response audio. Figure 2 summarizes the end-to-end pipeline. Unlike prior accent-adaptive systems that always apply accent conditioning, our approach introduces a confidence-gated personalization policy, where accent adaptation is applied only when the classifier confidence exceeds a threshold.

3.1. Accent estimation and caching

The system records the user’s speech through a microphone and predicts the speaker’s English accent using the trained classifier. The classifier is trained on three accents: Spanish-accented English, Korean-accented English, and Indian-accented English. The model outputs an accent label and confidence score, and caches the result for ten minutes to stabilize multi-turn interaction. The cached accent state is used for speech generation and triggers a safe fallback when the accent estimate is uncertain (Section 3.5).

The best-performing Self-Supervised Learning (SSL) accent classifier selected from our model comparison experiments; training details and quantitative results are described in Section 4.1 and Table 1.

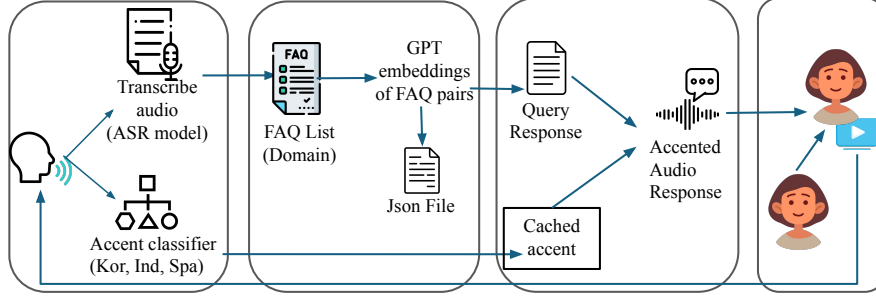


Figure 2: System flow for accent-adaptive FAQ response generation. The user’s audio is transcribed and classified for accent, closely aligned responses are retrieved from the validated FAQ set, and the grounded answer is rendered as accent-conditioned speech with a talking-face video.

3.2. Automatic speech recognition

We transcribe the input audio with the Whisper (small) model [16]. A low-latency ASR model is used to support task-oriented user queries and reduce processing time. The transcript and metadata (timestamp, predicted accent, confidence) are passed to the retrieval module.

3.3. FAQ-based response retrieval

The system retrieves responses to user queries from a curated FAQ set to eliminate the risk of unverified information. The ASR transcript is converted into a GPT-based text embedding [17] and compared with the precomputed embeddings of all FAQ questions. The FAQ entry with the highest cosine similarity is selected, and its paired answer is returned as a response. If retrieval confidence is low (e.g., the best similarity score is below a threshold), the system asks a clarification question instead of the wrong answer.

3.4. Accent-adaptive speech synthesis

The retrieved response text is converted to speech using VEVO TTS model [18], which supports reference-conditioned voice generation and enables accent-adaptive outputs from accented reference audio for each target accent (Indian, Korean, Spanish). This process uses (i) a session-level cached accent label from the accent estimator and (ii) reference-speech style representation to transfer accent cues and prosody while keeping the response content fixed.

We selected VEVO after preliminary trials with several recent reference conditioned TTS systems, including XTTS [19], F5-TTS [20], CosyVoice 2 [21], Vall-E [22], and MaskGCT [23]. Although these models generate fluent prompted speech, they struggle to maintain speaker identity and controlled accent transfer. In contrast, VEVO consistently generates desired accent features from the reference audio without damaging speaker timbre, hence, it was chosen as *LocalMATE* TTS backend.

We use a female-presenting voice for all responses, aligning with common design choices in widely deployed voice assistants. We do not explore gender-based personalization; instead, we focus on accent-adaptive speech output. This maintains a consistent system identity while adapting accent-related cues that may influence perceived understanding and trust.

3.5. Safe fallback for uncertain adaptation

Incorrect accent adaptation can reduce user confidence. To avoid this, we maintain a gating system based on the accent classifier’s confidence. When confidence falls below a threshold τ , the system defaults to a neutral US accented TTS output while preserving the response text. This approach ensures that accent adaptation is applied only when the estimated confidence is high, thus maintaining user trust.

3.6. Talking-face rendering

For audio-visual dialogue, the system generates a lip-synchronized talking-face video using SadTalker [24]. It takes the accented response audio and a portrait image that defines the systems visual identity as input and generates a synchronized talking-head video as the final response. The avatar is a photorealistic female avatar that is fixed to control the interface. This avoids demographic inference from speech and reduces appearance-related bias when evaluating accent-adaptive speech output.

4. Experimental Setup

4.1. Accent Classifier Setup

Various off-the-shelf speaker embedding models (e.g., ECAPA-TDNN classifiers, WavLM) can perform well on the datasets they are trained on, but their accuracy does not necessarily transfer to our target setting with Korean and Spanish accented English. Recent evidence in [25] shows that self-supervised speech encoders (e.g., Wav2Vec2/XLSR) capture accent-relevant phonetic variation more effectively than ECAPA-style embeddings and generalize better across accents when paired with simple utterance-level aggregation and lightweight augmentation. Additionally, prior studies show that accented TTS models can be finetuned with a limited amount of accented speech data (General American English, Irish English, and British English) [26]. Hence, we adopt an SSL-based approach and fine-tune a WavLM encoder with statistics pooling (mean+std) on a speaker-disjoint split of L2-ARCTIC for our three target accents (Indian, Korean, Spanish). This task-specific fine-tuning provides robust preprocessing for accent-adaptive response generation.

Data and splits. We train and evaluate the accent classifier on L2-ARCTIC [27] accented English speech, restricted to three accents (INDIAN, KOREAN, SPANISH). To evaluate model generalization to unseen speakers, we use a *speaker-disjoint*

Table 1: SSL accent classification results for Indian, Korean, and Spanish accents. XLSR denotes Wav2Vec2-Large-XLSR-53. Macro-F1 is the primary metric.

Model	Pooling	Dev F1	Test F1	Test Acc.
XLSR standard FT	Mean	0.547	0.271	26.0%
XLSR anti-overfit	Mask-aware mean	0.687	0.284	27.8%
WavLM-Base-Plus	Layer-weighted ASP	0.887	0.831	83.0%
WavLM-Base-Plus	Layer-weighted stats-avg	0.964	0.835	83.0%

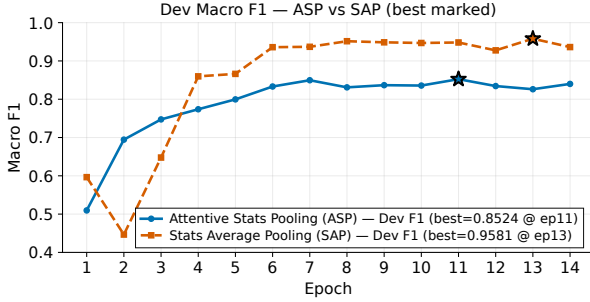


Figure 3: Dev Macro-F1 vs epoch for WavLM with ASP vs SAP pooling; best checkpoints marked (0.852@11 vs 0.958@13). SAP peaks higher and converges faster.

split with $N_{\text{train}}=6$ speakers for training (2 per accent), $N_{\text{dev}}=3$ speakers for development (1 per accent), and $N_{\text{test}}=3$ held-out speakers for testing (1 per accent), with zero speaker overlap across splits. The speaker IDs across the splits are: train = {Indian: ASI, RRBI; Korean: HJK, HKK; Spanish: ERMS, NJS}, dev = {Indian: SVBI; Korean: YDCK; Spanish: EBVS}, test = {Indian: TNI; Korean: YKWK; Spanish: MBMPS}. The splits contain 6,786/3,270/3,394 utterances for train/dev/test, respectively.

Input processing. All audio files are converted to mono and resampled to 16 kHz. For WavLM experiments, inputs are standardized to fixed-length 7.0 s segments (112,000 samples) via truncation or zero-padding. During training, we use random cropping for utterances longer than 7.0 s; during dev/test we use center cropping to reduce evaluation variance.

Model architecture. We fine-tune WavLM-Base-Plus [28] for 3-way accent classification. We output all hidden states and learn a softmax-normalized weighted sum over 13 states (embedding + 12 transformer layers). The resulting frame-level sequence is aggregated using statistics pooling (mean and standard deviation over time), producing a 1536-d utterance representation. A 2-layer MLP classifier (1536→256→3) with ReLU and dropout ($p=0.5$) predicts the accent label. The WavLM convolutional feature extractor is frozen throughout training.

Training. We optimize with AdamW using discriminative learning rates: 3×10^{-5} for the classification head and layer-weight parameters, and 5×10^{-6} for any unfrozen encoder parameters, with weight decay 0.02. We train for up to 50 epochs with batch size 4 and gradient accumulation of 2 steps (effective batch size 8), gradient clipping at 1.0, and label smoothing 0.05. To stabilize fine-tuning in a small, speaker-disjoint setting, we adopted gradual unfreezing [29], updating higher transformer layers before lower layers to reduce catastrophic forgetting.

In our implementation, the WavLM encoder was initially frozen while training the classification head and layer-

aggregation weights to establish a stable decision boundary. We then unfroze the last transformer layer at epoch 3 (near the end of the 4-epoch augmentation warm-up) to introduce limited encoder adaptation under mild perturbations. After training stabilized under stronger augmentation, we unfroze the last two transformer layers at epoch 6 (including the encoder layer normalization), increasing adaptation capacity while balancing generalization against overfitting. We select the best checkpoint based on development-set macro-F1 and apply early stopping with a patience of 6 epochs (minimum improvement 10^{-4}).

On-the-fly augmentation. Augmentations [30, 31] are applied during training only. We use stochastic gain perturbation (up to ± 3 dB), time shifting (up to 60 ms), additive Gaussian noise (std up to 0.003), and random bandpass equalization. Augmentation probability and strength follow a warmup+ramp schedule (warmup 4 epochs, ramp 10 epochs; probability 0.10→0.50 and strength 0.30→0.90).

Metrics. We report macro-F1 (primary) and accuracy on development and held-out test sets. Macro-F1 weights classes equally and is appropriate for speaker-disjoint evaluation where per-speaker variability can be large.

Compute and runtime. Experiments are run on a NVIDIA TITAN RTX GPUs (16 GB VRAM), using single-GPU training. For the WavLM setup, training time is approximately 291 s per epoch (~ 4.85 min/epoch), and the representative run shown converges with early stopping at 15 epochs (~ 73 min).

4.2. System Conditions

The *LocalMATE* system response generation dynamically adapts to the estimated user’s accent based on a set threshold. We evaluate it under four response conditions that differ only in the accent used for speech synthesis (other components, such as ASR, FAQ retrieval, and response content, remain unchanged).

Baseline (US Default) In the baseline condition, the system always produces a default U.S.-accented response, where no accent adaptation is applied.

Correct Accent Adaptation The system estimates participant’s accent using the fine-tuned classifier, and the response voice was experimentally controlled as EN-Korean to ensure a consistent realization of the intended correct-accent adaptation across all Korean-accented English participants.

Incorrect Accent Adaptation The system intentionally uses a mismatched accent that differs from the participant’s own accent (such as selecting a different accent than the one estimated for the participant) to simulate incorrect adaptation.

Uncertain + Fallback. When the accent classifier is uncertain (with confidence below a predefined threshold), the system uses the default U.S.-accented voice. This fallback avoids non-response under low-confidence cases and allows evaluation of user perceptions of confidence-gated fallback strategy.

4.3. Pilot User Study Setup

Before the main user study, we conducted a pilot experiment to validate the perceptual quality of the modified accent-conditioned speech and to ensure that synthesis artifacts did not confound the user study. In this online A/B preference pilot study, participants compared (*S*) and (*M*) stimuli. The modified versions were not disclosed to ensure user judgments were based on their perception of the audio and video. Our goal was to evaluate the comparability of our *Modified* outputs (*M*) against *Source* references (*S*) across accent, tone and overall quality, and to identify user modality preferences regarding the

relative advantage of audio versus video responses in agent interactions.

Design and stimuli. Stimuli were organized into three accent sections: *Set 1 (Indian)*, *Set 2 (Spanish)*, and *Set 3 (Korean)*, with each section including a male and a female speaker. For the audio-visual condition, *S* and *M* used the same avatar (one across male and female) to avoid confusing identity differences.

Participants and procedure. We recruited 327 participants through Microworkers and local outreach. Each of them was compensated 0.50 USD for a 5 minute task. Participants completed the study online using a structured web form with embedded audio/video stimuli. After applying consent, attention, and progression checks, a total of 111 valid submissions were available for analysis.

Measures. Participants rated (1–7 Likert; 4 = neutral) accent/tone match, naturalness, intelligibility, audio–visual synchronization (video), and preference for video over audio. We also collected an overall choice preference (*S* vs. *M*) and a one-sentence free-text justification.

4.4. Main User Study

We recruited 25 participants from the local university and surrounding community. We focused on Korean-accented English speakers as it is one of the system’s target accents and is required to evaluate accent alignment, mismatch, and confidence-gated fallback. Participants self-identified their English accent background and completed a brief spoken prompt as an initial step to confirm perceptible Korean-accented English. We removed participants without noticeable Korean-accented English or who failed the accent calibration (i.e., could not reliably distinguish the reference accents). We also excluded incomplete submissions and responses failing basic attention checks. Five participants were excluded, leaving $N=20$ participants for analysis (age range: 18–54). Each of them was compensated 5 USD for a 15 minute task.

Design and Tasks. We had each participant experience the four response conditions defined in Sec. 4.2 in a fixed order (*Baseline*, *Correct Accent Adaptation*, *Incorrect Accent Adaptation*, *Uncertain + Fallback*). The user study was conducted in-person in a quiet setting (≈ 15 min). Across conditions, ASR, retrieval, and response content were constant; only the response accent was updated. In each condition, participants interacted with the system vocally, posed two general airport FAQ-style questions (8 interactions total), and completed a brief post-trial survey. The questions covered common airport topics such as security checks, allowed items, baggage rules during layovers, lost passport procedures, bag drop, access for non-travelers, and power-bank allowance.

Procedure. After consent and eligibility checks, participants completed an audio/environment check and a short accent calibration task. They then completed the four conditions (Sec. 4.2). For each condition, participants spoke two queries, received the system’s accented spoken response with a synchronized talking-face video, and immediately completed the post-trial survey. The accent-adaptive logic used a confidence threshold of $\tau=0.34$: when the classifier confidence fell below τ , the system produced the fallback (U.S.-accented) response; in the mismatch condition, the system intentionally used a non-matching accent as specified in Sec. 4.2.

Measures. We collected objective interaction measures and subjective 7-point Likert ratings. Objective measures included repair behaviors (e.g., repetitions attempts). Subjective measures assessed perceived task success, interaction difficulty,

Table 2: *Accent calibration and eligibility from survey responses ($N=25$). Participants labeled three reference accent samples: Sample 1=Korean, Sample 2=Indian, and Sample 3=Spanish. We required at least 2/3 correct labels to proceed to the main study.*

Calibration item	Correct label	Accuracy (%)
Sample 1	Korean	80.0
Sample 2	Indian	80.0
Sample 3	Spanish	80.0
Distribution of number of correct labels out of 3		
0 correct	1 participant (4.0%)	
1 correct	4 participants (16.0%)	
2 correct	6 participants (24.0%)	
3 correct	14 participants (56.0%)	
Eligible (≥ 2 correct)	20/25 participants (80.0%)	
Excluded (< 2 correct)	5/25 participants (20.0%)	

confidence and trust, intention to rely, response quality, accent perception and match, and talking face quality (lip-sync and whether the talking-face video increased trust). In *Incorrect Accent Adaptation* and *Uncertain+Fallback*, participants also reported whether they perceived accent mismatch or system uncertainty and attributed any confidence loss to accent choice versus overall response quality.

To align measures with our research questions RQ1–RQ3, we used planned within-participant comparisons across the four trials: **A** (Baseline: default US accent), **B** (Correct Adaptation), **C** (Incorrect Adaptation), and **D** (Uncertain+Fallback).

RQ1 (Benefit of correct adaptation). We test whether correct accent adaptation improves user experience by comparing **B vs. A** on the primary subjective outcomes: *task success* (“I got the information I needed”), *interaction friction* (“I had to repeat myself”), *confidence* (“How confident are you in the system’s answer?”), *trust* (“I trust this system to provide correct airport information”), and *intention to rely* (“I would rely on this kiosk in a real airport”). Additionally, we report *overall response quality*, *effort to understand the voice*, and *audio naturalness*, and include video-specific ratings (*lip-sync/timing* and whether the *video increased trust*) to characterize the video-only response setting.

RQ2 (Cost of incorrect adaptation). We test the effect of incorrect adaptation by comparing **C vs. B** on the same primary outcomes (task success, friction, confidence, trust, reliance). To verify that the manipulation was perceived and to assess mechanism, we report (i) *perceived system accent* (“What accent do you think the system voice was?”) and *accent match* ratings, (ii) a mismatch detection item in Trial C (“Did the system voice accent feel mismatched for you?”), and (iii) confidence-loss attribution ratings (mismatch vs. overall response quality).

RQ3 (Fallback under uncertainty). We test whether fallback mitigates issues arising from accent estimation uncertainty by comparing **D vs. C** on the primary outcomes (task success, friction, confidence, trust, reliance). Moreover, we report Trial D follow-up questions from the *Fallback Evaluation* section: *reliability under uncertainty* and whether *fallback helped preserve confidence*, and checked perceived system uncertainty (“Did the system seem unsure about your accent?”).

Analysis. We compare the four system conditions using within-participant statistical tests and report condition-based

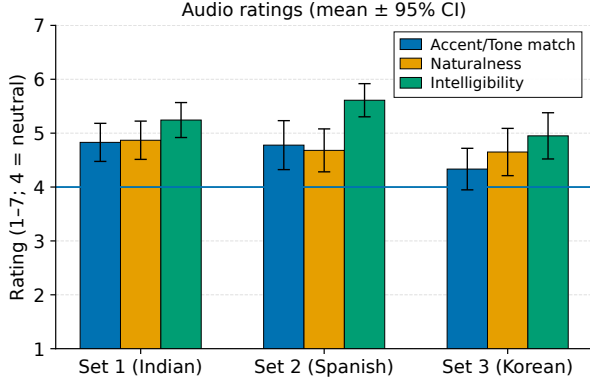


Figure 4: *Pilot study (audio-only): Mean participant ratings for the Modified (M) audio relative to the Source (S) reference, shown with 95% confidence intervals across accent blocks (Set 1: Indian, Set 2: Spanish, Set 3: Korean). Metrics are accent/tone match, naturalness, and intelligibility on a 1–7 Likert scale (4 = neutral).*

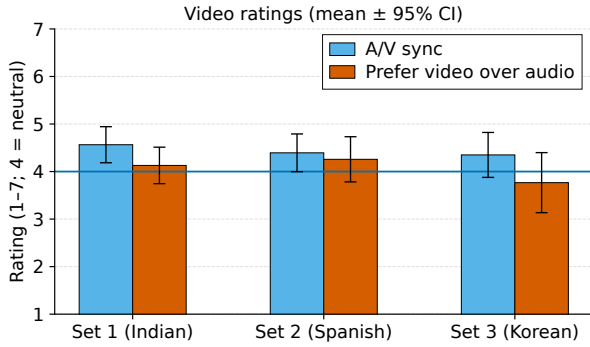


Figure 5: *Pilot study (audio-visual): Mean participant ratings with 95% confidence intervals across accent blocks (Set 1: Indian, Set 2: Spanish, Set 3: Korean). Metrics are perceived audio-visual synchronization and preference for video over audio (1–7 Likert; 4 = neutral).*

differences aligned with RQ1–RQ3 (B–A for RQ1, C–B for RQ2, and D–C for RQ3). For Likert-scale responses, we use non-parametric repeated-measures tests (Friedman followed by planned Wilcoxon signed-rank comparisons) with Holm correction for multiple comparisons, and we report effect sizes alongside p -values ($\alpha=0.05$). We analyze perceived system accent and accent-match ratings as manipulation checks to confirm that the intended accent differences were perceived.

5. Results

5.1. Accent Classification Results

Table 1 summarizes accent classification model performances on the held-out L2-ARCTIC test split. Wav2Vec2-based variants show weak held-out generalization (Test macro-F1: 0.271–0.284), despite moderate development scores, suggesting overfitting under limited speaker diversity. In contrast, WavLM-Base-Plus yields substantially stronger and more stable results across pooling strategies.

Although layer-weighted mean pooling achieves the high-

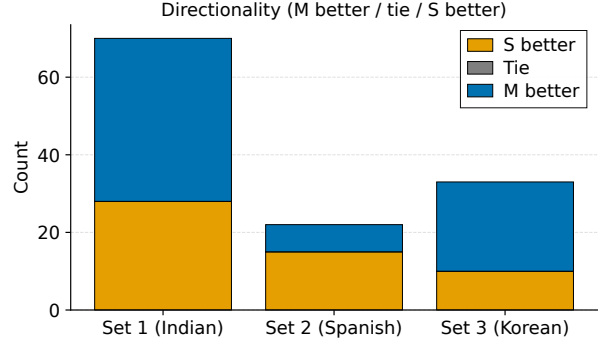


Figure 6: *Pilot study: General choice overall preference counts between Source (S) and Modified (M) across accent blocks (Set 1: Indian, Set 2: Spanish, Set 3: Korean).*

est held-out test macro-F1 (0.847), we deploy **statistics pooling** (layer-weighted + stats-avg; mean+std aggregation) due to its **robustness under distribution shift**. In our out-of-domain evaluation on 52 Korean-accented English utterances from the *Speech Accent Archive* [32], the statistics-pooling checkpoint achieves 92.3% accuracy (48/52), substantially outperforming the attentive-statistics pooling checkpoint (63.5%, 33/52). Given that accent predictions are used as a control signal for downstream generation and fallback gating, we prioritize stability and transfer over small differences on the in-domain test split.

Residual errors are dominated by Korean–Spanish confusions under distribution shift, consistent with phonetic overlap

5.2. Pilot Study Results

Prior to the main user study, we validated stimulus quality by running a perceptual pilot that compared *Source* (S) and *Modified* (M) outputs across three accent blocks (Set 1: Indian, Set 2: Spanish, Set 3: Korean). Across blocks, *M* was rated above the neutral midpoint on audio-only quality measures (Figure. 4), with the highest scores on intelligibility and naturalness. Mean intelligibility ranged from 5.38–6.00 and naturalness ranged from 4.82–5.11 across sets, while accent/tone match was slightly above neutral (4.76–4.91). These results suggest that the modified synthesis mostly retained the target speaking style.

Video-specific ratings were closer to neutral (Figure. 5). Audio-visual synchronization ranged from 4.21–4.71, and preference for video over audio ranged from 4.20–4.30. This suggests that adding a talking-face video did not reliably improve perceived quality in this pilot.

The general preference for audio and video (Figure. 6) differed across accent blocks. Participants preferred *M* for Set 1 (Indian; 33/56) and Set 3 (Korean; 23/33), but preferred *S* for Set 2 (Spanish; 15/22). This pattern could mean that the Spanish block was more sensitive to synthesis artifacts or speaker/style mismatch. In open-ended explanations, participants most often preferred clarity of understanding, naturalness, and overall audio quality.

5.3. Main User Study Results

We analyze within-participant differences across four response conditions (**A**=Baseline default US accented voice, **B**=Adaptive-Correct, **C**=Adaptive-Mismatched, **D**=Fallback

Table 3: Main user-study outcomes across four within-participant conditions ($N=20$): **A**=Baseline (default US voice), **B**=Adaptive-Correct, **C**=Adaptive-Mismatched, and **D**=Fallback (uncertain $\rightarrow$ default). We report per-condition medians (med), Friedman tests across A–D, and planned Wilcoxon signed-rank comparisons aligned with RQ1–RQ3. Planned-comparison cells report Holm-corrected p_h and effect size $r = |Z|/\sqrt{n}$. For friction, lower values are better.

Outcome	A med	B med	C med	D med	$\chi^2(3)$	p	RQ1: B–A	RQ2: C–B	RQ3: D–C
Confidence	5.5	6.0	4.0	7.0	28.30	< .001	.026/.703	.002/.841	< .001/.886
Trust	6.0	6.5	4.0	7.0	21.49	< .001	.032/.677	.001/.828	.001/.820
Friction (repeat required)	3.5	2.0	4.0	3.0	12.29	.006	.031/.650	.016/.675	.017/.660

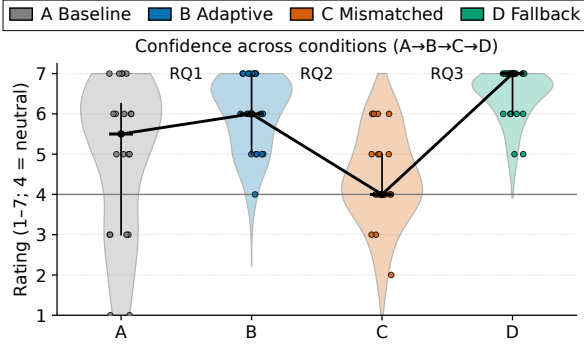


Figure 7: Confidence ratings (1–7) across A–D.

under uncertainty). Survey data were collected after each condition ($N=20$). We visualize outcome distributions with graph plots and individual points (Figure. 7–9) and report condition-level Friedman tests followed by planned within-participant comparisons aligned with our research questions (RQ1: B–A; RQ2: C–B; RQ3: D–C). We use Wilcoxon signed-rank tests with Holm correction (per outcome across the three planned tests) for planned comparisons and report effect sizes $r = |Z|/\sqrt{n}$, where n is the number of non-zero paired differences.

Manipulation check (perceived accent). Our hypotheses depend on perceived accent differences, hence, we first verified that the system’s reference accents were detectable. Participants completed an accent calibration (Sample 1=Korean, Sample 2=Indian, Sample 3=Spanish) and only those with $\geq 2/3$ correct were allowed to proceed. Calibration performance was high (80% accuracy on each sample; 20/25 eligible), and all participants reported that they could clearly tell differences between the accents (Table 2). This indicates the accent manipulation was perceptible for our set of participants.

RQ1 (B vs A): Correct accent adaptation improves user confidence/trust and reduces repeated user querying. Correct adaptation (B) improved participants’ experience relative to the baseline (A). The users’ perception shows an upward shift from A to B for confidence and trust (Figure. 7, 10) and a downward shift for friction (Fig. 8; lower is better). The Wilcoxon comparisons (Holm-corrected) confirm higher confidence ($p_h=.026$, $r=.703$), higher trust ($p_h=.032$, $r=.677$), and lower friction ($p_h=.031$, $r=.650$) in B than A (Table 3).

RQ2 (C vs B): Incorrect adaptation reverses the gains and is experienced as mismatch. Mismatched adaptation (C) produced a clear degradation relative to the correct adaptation (B). As shown in Figure. 7–8, confidence and trust decline from B to C while friction increases. Moreover, the comparisons show lower confidence ($p_h=.002$, $r=.841$), lower trust ($p_h=.001$, $r=.828$), and higher friction ($p_h=.016$, $r=.675$) in C than B

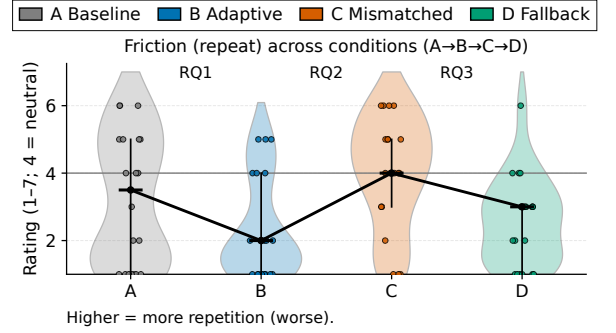


Figure 8: Friction (repeat required) ratings (1–7; higher = worse) across A–D.

(Table 3). These results indicate that committing to an incorrect accent choice can be more harmful than maintaining a consistent default voice.

RQ3 (D vs C): Confidence-gated fallback mitigates mismatch harms under uncertainty. The fallback under uncertainty (D) mitigated the negative effects of the mismatch observed in C. Confidence and trust increase from C to D and friction decreases (Figure. 7–8). The comparisons show higher confidence ($p_h<.001$, $r=.886$) and higher trust ($p_h=.001$, $r=.820$) under fallback, along with lower friction ($p_h=.017$, $r=.660$), compared to mismatched adaptation (Table 3). These results support confidence-gated fallback as a safer strategy than always committing to an accent prediction.

Video-based trust (secondary). The participants’ ratings for the talking-face video increased trust broadly follow the same pattern as the primary outcomes: improved ratings under correct adaptation (A $\rightarrow$ B), degraded ratings under mismatch (B $\rightarrow$ C), and partial recovery under fallback (C $\rightarrow$ D) (Fig. 9).

Across all outcomes, we observe a consistent pattern: A $\rightarrow$ B improvement, B $\rightarrow$ C degradation, and C $\rightarrow$ D recovery. This indicates that correct accent adaptation improves interaction quality, while incorrect adaptation harms user trust. The confidence-gated fallback effectively mitigates this harm. (Table 3; Figure. 7–8). In conclusion, these results support confidence-gated accent adaptation; adapting when confident improves user experience, while falling back under uncertainty avoids the larger costs of incorrect adaptation.

6. Discussion

We designed *LocalMATE* to evaluate how accent-adaptive response generation influences user experience in a spoken airport kiosk setting. This included cases of incorrect accent adaptation and low-confidence predictions. Evaluations show accent

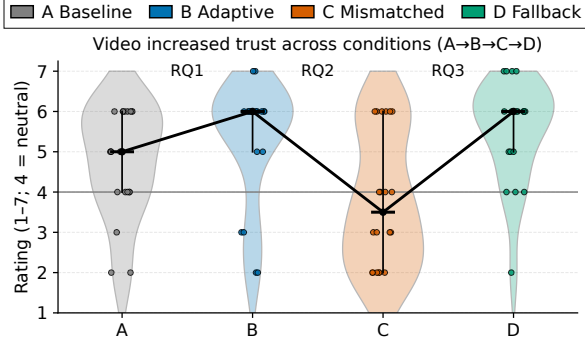


Figure 9: Video based trust ratings (1–7) across A–D.

enhances comfort and trust, but misalignment can reduce user confidence. We interpret these outcomes in relation to RQ1–RQ3 and discuss the impacts of deploying accent-adaptive systems in public systems.

RQ1: Why correct accent adaptation helps. Our results imply that perceived accent alignment enhances interaction naturalness and system competence, serving as a social cue that reduces perceived inconvenience in brief task-oriented interactions. For public kiosks, where interactions are brief and users are often short of time, such social and perceptual cues may notably influence willingness to rely on the system.

RQ2: Why incorrect adaptation can be expensive. We observed that incorrect accent adaptation leads to user recognition failure. This creates a noticeable social mismatch compared to standard TTS responses. This leads to a major trade-off; while personalization is beneficial, an incorrect accent can reduce trust, even with accurate content. Hence, accent detection accuracy should be prioritized over increasing coverage to maintain a reliable system.

RQ3: Why confidence-gated fallback is a practical safeguard. The fallback condition defaults to a neutral American accent unless the accent classifier has a high confidence threshold, reducing accent adaption errors while preserving service availability. However, fallback should be viewed as a short-term intelligibility safeguard rather than a replacement for accent adaptation, since longer interactions may make accent switching more noticeable and affect perceived identity, similarity, trust, and voice stability [33]. Thus, ‘safe personalization’ requires both a robust classifier and a reliability-first decision policy under uncertainty.

Why robustness in accent prediction matters for end-to-end systems. The accent estimator is the first step in the accent-adaptive pipeline, so misclassifications become user-visible failures. Robustness and calibration are essential: over-confidence can bypass the fallback gate and trigger incorrect personalization. Accent adaptation should therefore be treated as a critical control signal, with misclassifications tightly managed to preserve system integrity.

Role of talking-face responses in kiosk interactions. The pilot study suggests that accent-conditioned speech mainly improves perceived response quality. Intelligibility and naturalness were rated above neutral across accents, while video-specific ratings (audio–visual sync and preference for video over audio) were closer to neutral, indicating the talking face was acceptable but not essential. However, embodiment may

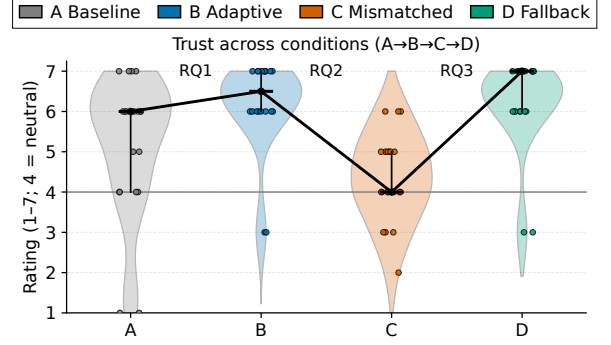


Figure 10: Trust ratings (1–7) across A–D.

enhance presence and interaction in public, task-oriented settings, motivating our evaluation of *LocalMATE*.

Implication for Public Voice Agents. These findings suggest that personalization in voice interfaces must be deployed conservatively, as incorrect adaptations can harm user trust more than a neutral voice. Confidence-aware personalization are essential for real-world deployment.

Limitations and future work Our main study focuses on Korean-accented English speakers, which supports controlled evaluation but limits generalization to other accents and multi-lingual settings. Interactions were short (two queries per condition) in a brief, in-person session, however, longer multi-turn use may lead to additional repair behavior and changes in trust over time. The fixed condition order may have introduced order effects, so future studies should address this through randomized ordering. Accent strength varied across participants but was not externally assessed, and the controlled evaluation setting may not fully capture real-world noise, device variability, or their effects on ASR, accent estimation, and perceived response quality.

First, we plan to expand to more accents, personalize adaptation based on accent strength and user preference, and improve accent-confidence calibration under real-world conditions using uncertainty-aware policies such as dynamic thresholds. In the future, we also plan longer, more realistic multi-turn evaluations and will compare audio-only versus audio+video responses across contexts.

7. Conclusion

We present *LocalMATE*, an accent-adaptive response system that uses a three-class accent estimator (Indian, Spanish, Korean) and a confidence threshold to fall back to a neutral US accent voice when predictions are uncertain. WavLM-Base-Plus with statistics (mean+std) pooling delivered the most reliable accent classification for deployment and generalized well on external Korean-accented English from the Speech Accent Archive. In a perceptual pilot comparing *Source* (S) and *Modified* (M) outputs, modified audio was rated above neutral for intelligibility and naturalness, while talking-face ratings remained closer to neutral and preferences varied by accent (with Spanish modified outputs less preferred). In the end-to-end user study, correct accent adaptation improved trust, confidence, and task outcomes, whereas incorrect adaptation reduced confidence; the confidence-based fallback mitigated these negative effects by avoiding adaptation when the system was uncertain.

8. Generative AI Use Disclosure

The authors declare that no generative AI was used in the creation of the manuscript.

9. Acknowledgement

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